

8	C	Force constant, $k = \frac{5.0}{0.1} = 50 \text{ N m}^{-1}$ Initial extension $= \frac{F}{k} = \frac{3}{50} = 0.06 \text{ m}$ and final extension $= \frac{F}{k} = \frac{2.5}{50} = 0.05 \text{ m}$ Change in E.P.E. $= \frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2 = \frac{1}{2} (50)(0.05)^2 - \frac{1}{2} (50)(0.06)^2 = -0.028 \text{ J}$
9	D	Frictional force on the object provides the centripetal force $mr\omega^2$. Both objects have the